

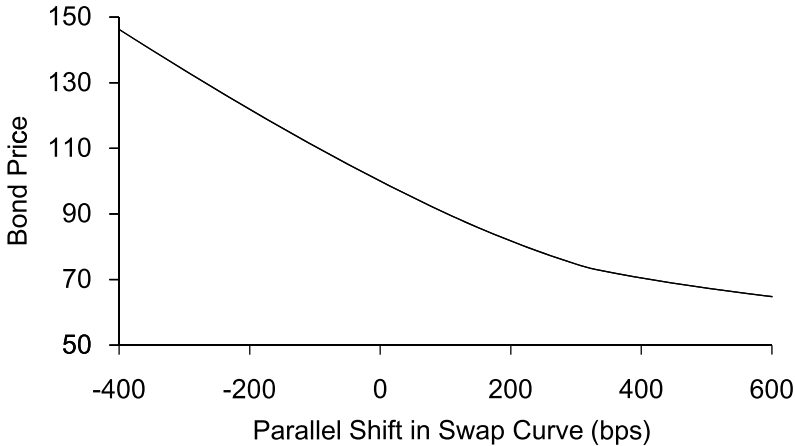


Figure 7.6: Price Sensitivity of the Leveraged Inverse Floater Given Instantaneous Parallel Shift in the Swap Curve

value of the bond when the swap curve is not shifted. The duration of the leveraged inverse floater based on Equation 7.5 is shown in Figure 7.7.

A striking result from this figure is that this leveraged inverse floater, which has a final maturity of 4 years, has a duration of nearly 10 in the current swap curve environment! Most bonds have a duration less than their maturity, but this is not the case for the leveraged inverse floater. The magnitude of the duration indicates that the value of this bond is extremely sensitive to movements in interest rates. The fact that the duration is positive indicates that as the market rallies, the value of the leverage inverse floater goes up, and as the market sells off, the value of the leverage inverse floater goes down. But it is the magnitude of the duration that illustrates just how incredibly sensitive this bond is to movements in interest rates.

### Decomposing the Leveraged Inverse Floater

Why does the leverage inverse floater have such a high duration? In order to understand this, let's take a look at the two portfolios in Table 7.9. Portfolio A contains a long position of \$100mm of the leveraged inverse floater that we just structured. Portfolio B contains a long position of \$300mm of fixed rate bonds with a coupon of 16.34%/3, a short position of \$200mm LIBOR floaters, and a long position of \$200mm caps struck at 16.34%/2. These two

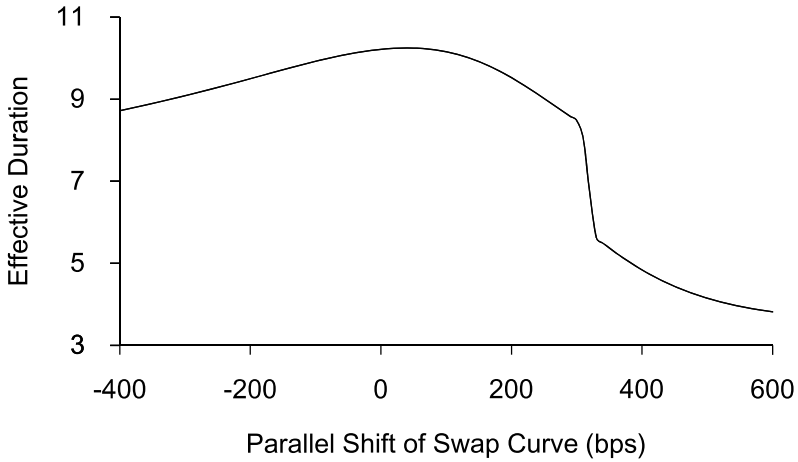


Figure 7.7: Effective Duration of the Leveraged Inverse Floater

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Portfolio A:

Long \$100mm of 16.34% – 2 × 1-year LIBOR Leveraged Inverse Floater

Portfolio B:

Long \$300mm Fixed Rate Bonds that Have a Coupon of 16.34%/3

Short \$200mm LIBOR Floaters

Long \$200mm Caps Struck at 16.34%/2

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Table 7.9: Decomposition of Leveraged Inverse Floater

portfolios are in fact identical from risk and return perspectives. To show that this is true, first note that each portfolio has a long position in \$100mm of principal. This is obviously the case for Portfolio A. Portfolio B is long \$300mm fixed rate bonds and short \$200mm LIBOR floaters, which is a net long of \$100mm of principal as well. Now let's show that each portfolio generates the same coupon flows. Portfolio A, the leveraged inverse floater, pays coupon interest equal to

$$\$100\text{mm} \times \max(16.34\% - 2 \times \text{LIBOR}, 0). \quad (7.6)$$

Portfolio B pays coupon interest equal to

$$\begin{aligned}
& \$300\text{mm} \times 16.34\%/3 - \$200\text{mm} \times \text{LIBOR} + \\
& \$200\text{mm} \times \max(\text{LIBOR} - 16.34\%/2, 0) \\
& = \$100\text{mm} \times 16.34\% - \$100\text{mm} \times 2 \times \text{LIBOR} + \\
& \quad \$100\text{mm} \times \max(2 \times \text{LIBOR} - 16.34\%, 0) \\
& = \$100\text{mm} \times (16.34\% - 2 \times \text{LIBOR}) + \\
& \quad \$100\text{mm} \times \max(2 \times \text{LIBOR} - 16.34\%, 0) \\
& = \$100\text{mm} \times [(16.34\% - 2 \times \text{LIBOR}) + \max(2 \times \text{LIBOR} - 16.34\%, 0)] \\
& = \$100\text{mm} \times \max(16.34\% - 2 \times \text{LIBOR}, 0),
\end{aligned}$$

which is the same as Equation 7.6. Thus, the two portfolios are the same. But this gives us some insight as to why the leveraged inverse floater has such a high duration. Let's focus on Portfolio B now, which is long \$300mm fixed rate bonds and short \$200mm LIBOR floaters (and also long some caps, which we will ignore for now). So it's sort of like Portfolio B consists of a long position of three fixed rate bonds of \$100mm principal each and short two floating rate bonds of \$100mm each. What is being accomplished in this portfolio is that a leveraged long position is being established. That is, \$200mm LIBOR floaters (which have near zero duration) are being sold to help finance the purchase of \$300mm of the fixed rate bonds (each of which has positive duration). The net result is a long leveraged position that has a very high duration. The buyer of this inverse floater is essentially buying three fixed rate bonds (and selling two floaters to partially finance the position), and thus the duration of \$100mm of the inverse floater is similar to the duration of \$300mm of fixed rate bonds.<sup>22</sup>

By the way, notice in Figure 7.7 that when the market sells off dramatically, the duration of the inverse floater begins to decline. When rates are very high, the market's expectation of all future coupons paid by the inverse floater is zero. Thus, the risk of the inverse floater resembles that of a zero coupon bond in high rate scenarios. And zero coupon bonds have modified durations that are slightly less than their maturities.<sup>23</sup>

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<sup>22</sup>The caps in Portfolio B should also be accounted for when we are considering the duration of the portfolio. As interest rates fall, the caps will lose value, and thus their existence will reduce the duration of Portfolio B. But these high-strike caps will have a relatively small effect in terms of the overall price sensitivity of the portfolio in a low interest rate environment. However, in a high interest rate environment, their price sensitivity is significant.

<sup>23</sup>Recall in our decomposition of the inverse floater we said that an inverse floater is equivalent to a long position in fixed rate bonds, a short position in floaters, and a long position in caps. In examining the duration of the inverse floater in the current yield curve environment, we first ignored the caps. That is a reasonable thing to do when the inverse floater is first priced, and the cap strike is relatively high. In this case the sensitivity of

## 7.7.2 Orange County

Orange County, a municipality located in Southern California, came into the spotlight in 1994 when it announced bankruptcy in December of that year. The bankruptcy was the result of the mismanagement of its investment portfolio that invested in a variety of fixed income securities, most notably structured notes. The county's \$7.5bn investment portfolio had staggering losses totaling approximately \$1.7bn.

In particular, Orange County was known to have invested heavily in inverse floaters. There is nothing inherently wrong with inverse floaters, or their high-octane cousins, leveraged inverse floaters. But in the case of Orange County, the combination of inverse floating rate notes and leverage and a Fed tightening cycle combined to bring about the bankruptcy of the county's investment portfolio.<sup>24</sup> How much the county's treasurer, Bob Citron, understood about what he was buying is a matter of some debate.<sup>25</sup>

Technically, it wasn't derivatives that brought about the bankruptcy of Orange County. It was repo. But perhaps the difference is semantics. The essential strategy of Orange County, as we will discuss in some detail, was to borrow and to make leveraged bets that were long the market. The technology to implement leverage within a portfolio has been around a long time and certainly doesn't require derivatives. But the structured note market did make it convenient to make very highly leveraged bets on interest rates (i.e., that rates would remain low). This strategy worked out remarkably well when rates were low. But when the Fed went on a protracted mission of raising interest rates in 1994 (the Fed raised rates six times in that year), the results were disastrous.

Orange County added to its risk from inverse floaters by adding leverage via repo. It entered into repo transactions with the very same dealers who sold the county these bonds so that it would have more money to buy even more bonds. For example, when Orange County bought a leveraged inverse

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the cap to changes in interest rates is small. But as the market sells off, the interest rate sensitivity of the caps increases significantly.

<sup>24</sup>This is not to suggest that Orange County only purchased inverse floaters. In his book *Big Bets Gone Bad*, Jorion (1995) states that around the time of the bankruptcy the Orange County investment portfolio was composed of around 60% of structured notes, although there were positions in other securities such as mortgages and corporates.

<sup>25</sup>Jorion (1995) reports that when Citron was asked in 1993 why he believed that interest rates would remain low, he responded, "I am one of the largest investors in America. I know these things." But in 1994, the Fed started raising rates. Jorion quotes then-Assistant Treasurer of Orange County Matthew Raabe as saying that Citron was in "a state of denial over this," referring to the troubles of the county's investment portfolio. Jorion reports that Raabe claims that Citron began consulting a psychic and a mail-order astrologer at work.

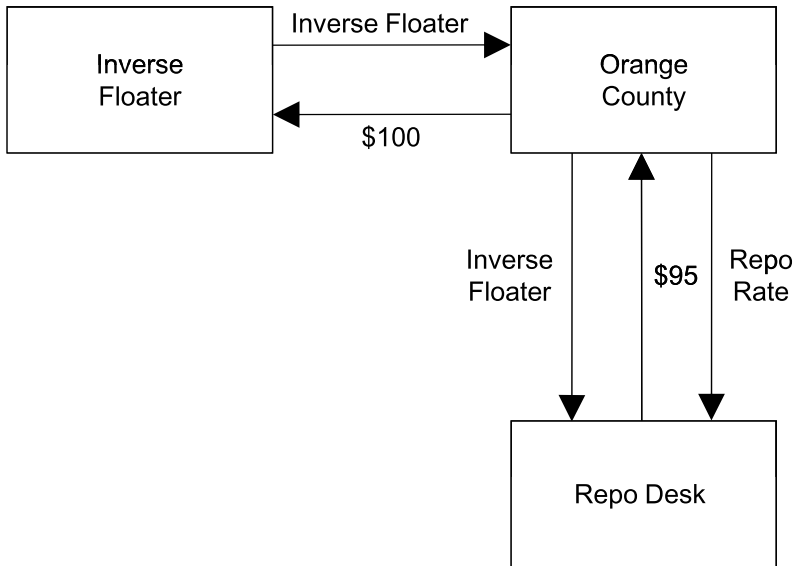


Figure 7.8: Orange County Financed Its Purchase of Levered Inverse Floaters

floaters (which is a leveraged long position in and of itself), it would often finance the bond (creating even more leverage) so that it could buy even more inverse floating rate notes.

This arrangement is demonstrated in Figure 7.8. In this figure, Orange County purchases \$100mm of an inverse floater. In order to finance the purchase, Orange County enters into a repo agreement with the repo desk of the dealer who sold Orange County the bond. In the repo, which might last typically for some number of days or weeks or months, Orange County pledges the bond as collateral on a loan and pays some rate of interest, known as the repo rate. Typically, there is a *haircut* that is applicable, which stipulates that Orange County will not receive the full value of the pledged asset but only a portion of it. In this case, we are assuming a 5% haircut, meaning that Orange County was able to finance \$95mm of the \$100mm purchase price. Moreover, and very important, the terms of the repo would stipulate that as the value of the bond fell, Orange County would be required to post more collateral to the repo desk. Thus, if the bond was purchased today at par, and tomorrow it was worth only \$98mm, Orange County was responsible for posting an additional \$2mm of collateral to the repo desk. In the meantime, Orange County could go ahead and purchase more bonds with the \$95mm cash it received via the repo transaction. In other words,

\$100mm of leveraged inverse floaters could be purchased by actually putting up only \$5mm.

Between 1992 and 1993 when the Fed was on hold and rates were stable, the coupon interest that Orange County earned on its bonds more than offset the repo interest it paid to finance some of its purchases. But when rates shot up in 1994, it was a very different story.

In 1994, the Fed became very concerned about the risk of inflation and raised the Fed Funds rate from 3% to 5.5%. The bond market sold off tremendously, and of course LIBOR rates also rose. This served to diminish the value of fixed rate bonds. Leveraged inverse floaters were particularly hard-hit due to their extreme sensitivity to changes in interest rates (refer again to Figure 7.7). As rates rose, this high duration eroded the value of the inverse floaters that Orange County had repoed. Under the terms of the repo, Orange County was required to deliver cash to the dealer when the value of the bond went down. At the same time, with LIBOR rates going up over the course of the year, the coupon interest earned on the inverse floaters was going down. And to make matters still worse, the repo rate Orange County would face on any additional repos would increase with the general rise in rates. It was the perfect storm.

As rates first started to rise, Orange County initially had enough cash on hand to satisfy its obligations. But as time went on, and as the Fed continued to raise rates and the market sold off further, Orange County had to post more and more collateral under its repo agreements. According to Jorion (1995), Orange County expended around \$515mm between February 1994 and May 1994 to satisfy the collateral calls from its repo obligations. Between August and November of that same year, Jorion's analysis shows that the cash balance of the fund fell around \$1bn. Shortly thereafter, Orange County declared bankruptcy as banks started liquidating the collateral they held when Orange County was unable to meet its margin requirements.

This is a dramatic simplification of the dynamics at work that brought on the ultimate bankruptcy of Orange County. But it contains the salient features of Orange County's bankruptcy and the ultimate liquidation of its investment portfolio.<sup>26</sup> Recall that in Figure 2.2 of Chapter 2, we pointed out that there was a period of time at the end of 1994 when 2-year swap spreads gapped up dramatically when it was announced that Orange County had gone bankrupt, and liquidation of its investment portfolio seemed imminent. The market was aware that Orange County had purchased a great many inverse floaters and that when it came time to liquidate, it would have to

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<sup>26</sup>Orange County hired the investment bank Salomon Brothers to help advise on the portfolio, which eventually conducted a liquidation of its portfolio of structured notes. Dealers bid on many of the bonds in an auction.

sell these bonds. Any dealer who bought inverse floaters from Orange County during the liquidation would be getting long the market. In order to hedge this position, swap dealers would need to short the market, which they often do by paying fixed in swaps. It was also well known in the market that most of these bonds in Orange County's portfolio had a few years left to maturity. Thus, dealers who bought the bonds would have to pay fixed in swaps whose maturity matched the maturity of these bonds. This explains why 2-year swap spreads moved so dramatically during this time. Orange County was not known to have purchased any leveraged inverse floaters maturing in, say, 10 years. Thus, we did not see a similar move in 10-year spreads.

## 7.8 Range Notes

Range notes were fairly exotic structures when they came on the scene in 1993. There are several different versions of range notes that have been popular over the years (such as binary accrual notes and LIBOR Enhancement Accrual Notes [LEANs]). The basic structure of these bonds is that they pay coupons of the following form:

$$\text{Daily Coupon Accrual} = \begin{cases} C\% & \text{when Index is within Range} \\ Y\% & \text{when Index is outside Range.} \end{cases}$$

The coupon  $C$  is an above market coupon, and  $Y$  is a below market coupon, often zero. The underlying Index is often 3-month LIBOR. The Range is set at the beginning of the trade. For example, consider a range note written on 3-month LIBOR and a Range of 3% to 4.25%. Provided that 3-month LIBOR remains in the Range of 3% to 4.25%, an investor who owns the bond earns the high coupon  $C$  and otherwise earns the low coupon  $Y$ .

### 7.8.1 LEANs

The first type of range note that we will look at is the LEAN. This is a floating rate note that pays a very high coupon when the underlying index LIBOR is in a range and a low coupon (often zero) when LIBOR sets out of the range. The terms of such a bond are shown in Table 7.10. The margin  $x$  was typically much higher than what one could obtain on comparable floating rate notes. One thing to note in the terms of the LEAN in Table 7.10 is how LIBOR is used to determine coupon resets versus how LIBOR is used to determine how many days each quarter the nonzero coupon accrues. To take a particular example, suppose that on a coupon reset date, 3-month LIBOR is equal to 4.00% and suppose that the margin  $x$  is equal to 0.50% for the trade. Then for that quarter the maximum possible coupon that will accrue

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|              |                                                                                                                                                                   |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Issuer:      | U.S. Agency                                                                                                                                                       |
| Principal:   | \$100mm                                                                                                                                                           |
| Maturity:    | 3 years                                                                                                                                                           |
| Coupon:      | 3-month LIBOR + $x$ each day LIBOR is between 3% and 4.25%<br>0% otherwise                                                                                        |
| Day Count:   | Act/360                                                                                                                                                           |
| Resets:      | Quarterly for the determination of the Coupon that is to be accrued<br>Daily for purposes of calculating the number of days in a quarter<br>the coupon is accrued |
| Payments:    | Quarterly                                                                                                                                                         |
| Issue Price: | 100%                                                                                                                                                              |

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Table 7.10: Terms of a LIBOR Enhancement Accrual Note

each day is 4.50%. During the quarter, 3-month LIBOR is observed daily for the purposes of determining accrual. Denote by  $N$  the number of days during the quarter that 3-month LIBOR is in the range of 3% to 4.25% and denote by  $M$  the actual number of days in the quarter. Then the payment that the bondholder will receive for the quarter is

$$4.50\% \times \$100\text{mm} \times N/M \times \text{Act}/360.$$

LEANs first came into existence in January 1994. According to Peng and Dattatreya (1995), within a two-week period, over \$1bn of new issue LEANs had been created. Investors rushed in to buy LEANs because of the low level of LIBOR volatility that had recently been observed and the belief that there would not be any rate hikes to push LIBOR out of the range. As shown in Figure 7.9, LIBOR had been setting in a very tight range for the prior two years. Some people who looked at a picture like the one in Figure 7.9 thought LEANs were a pretty good bet. But in early 1994, the Fed became concerned about inflationary pressures and raised short-term rates a total of six times that year (the Fed Funds rate went from 3% to 5.5% over the course of that year). This pushed LIBOR up significantly, as seen in Figure 7.1, which pushed many LEANs outside of their accrual ranges. Thus, instead of getting above market coupons from these bonds, many of these bonds started paying zero coupons. Moreover, based on high forward rates, most pricing models placed very little value on future coupons on these range notes. These bonds, which had been bought at par, were now trading more like zero coupon bonds.



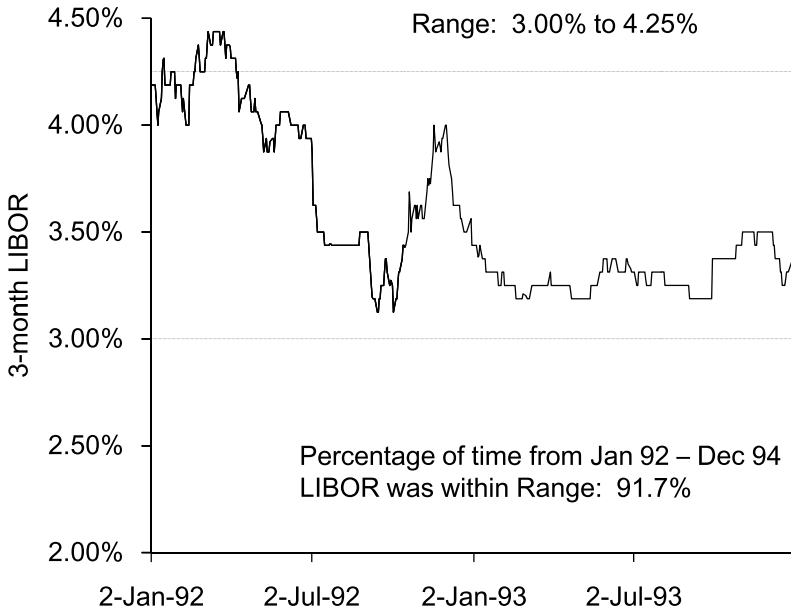


Figure 7.9: Historical LIBOR in the Few Years Preceding the Existence of LEANs

*Source:* Figure adapted from Peng and Dattatreya (1995).

## 7.8.2 Binary Accrual Notes

Another popular type of range bond (which has been more popular in recent years) is the binary accrual note. These are similar to LEANs except the coupons paid, provided the index remains in the predetermined range, are fixed. As a particular example, consider the terms of a binary accrual note as shown in Table 7.11.

### Example: Pricing a Binary Accrual Note

We will price up a 4-year binary accrual note in the context of our stylized example in which swaps pay annual bond versus 1-year LIBOR. Assume the swap curve is as in Table 7.5. Suppose that an investor wants to purchase a binary accrual note that pays a coupon of  $C\%$  if 1-year LIBOR is between 4% and 6.5% and zero otherwise. We will make the simplifying assumption that 1-year LIBOR is observed only once at the beginning of each year to determine whether the investor will receive a coupon of  $C\%$  or 0%. Assume

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|              |                                                                                            |
|--------------|--------------------------------------------------------------------------------------------|
| Issuer:      | U.S. Agency                                                                                |
| Principal:   | \$100mm                                                                                    |
| Maturity:    | 3 years                                                                                    |
| Coupon:      | 6.00% each day LIBOR is between 3% and 4.25%<br>0% otherwise                               |
| Day Count:   | Act/360                                                                                    |
| Resets:      | Daily for purposes of calculating the number of days in a<br>quarter the Coupon is accrued |
| Payments:    | Quarterly                                                                                  |
| Issue Price: | 100%                                                                                       |

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Table 7.11: Terms of a Binary Accrual Note

the issuer pays LIBOR flat. What level of  $C$  can we provide to the investor?

The coupon paid to the investor each year as a function of LIBOR is shown in Figure 7.10. This payout diagram is referred to as a *binary corridor*.<sup>27</sup> For the payment in the binary accrual note that occurs in one year based off of today's LIBOR set, there is no uncertainty. The coupon payment in one year will be  $C\%$  since today's LIBOR setting of 5% falls inside the range of 4% to 6.5%. The remaining coupon payments at time  $t_{i+1}$ , where  $i = 1, 2, 3$ , are uncertain.

Under the usual Black assumptions, the value of a binary option that pays out  $C\%$  at time  $t_{i+1}$  if LIBOR at time  $t_i$  is greater than  $X$  and zero otherwise is

$$\frac{1}{[1 + \text{zc}(t_{i+1})]^{t_{i+1}}} \times C\% \times N(d_2[X]), \quad (7.7)$$

where

$$d_2[X] = \frac{\ln[f(t_i, t_{i+1})/X] - \sigma_{t_i}^2 \times t_i/2}{\sigma_{t_i} \times \sqrt{t_i}}. \quad (7.8)$$

In Equations 7.7 and 7.8, we have accentuated that fact that  $d_2$  depends on the strike  $X$  by employing the notation  $d_2[X]$ . Using Equation 7.7 and 7.8, the value of the binary corridor shown in Figure 7.10 is

$$\frac{1}{[1 + \text{zc}(t_{i+1})]^{t_{i+1}}} \times C\% \times [N(d_2[X_1]) - N(d_2[X_2])], \quad (7.9)$$

where  $X_1$  denotes the lower boundary of 4%, and  $X_2$  denotes the upper boundary of 6.5%.

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<sup>27</sup>For a review of binary options, see Appendix A.